Helena Fu

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Education

Tufts University Medford, MA
B.S. Computer Science, B.A. Mathematics, Minor: AI Development *Jan 2027*
(GPA: 3.92 / 4.00)
Honors: 4x Tufts Dean's List, Al Gardner Memorial Endowment Scholarship (Society of Women Engineers)
Relevant Coursework: Algorithms, Data Structures, Database Systems, Operating Systems, Intro to Security, Web Programming, Machine Learning, Mathematical Modeling, Linear Algebra, Machine Architecture and Assembly

Technical Skills

Languages: JavaScript, TypeScript, Python, Java, C++, SQL, HTML/CSS, Bash
Frontend/Backend: React, Next.js, Tailwind CSS, WordPress, Node.js, Flask, Express, REST APIs, MongoDB, SQL
DevOps/Cloud: Docker, AWS, Google Cloud Platform, Palantir Foundry
Developer Tools: Git/GitHub, Jira, Asana, Tableau
Additional: Scikit-learn, PyTorch, TensorFlow, HuggingFace

Experience

Mad Realities New York City, NY
Software Engineering Intern *June 2025 – Aug. 2025*
- Developed a RAG-based SMS chatbot with Cloudflare, Flask, Docker, and Community webhooks, engaging 3,000+ users and boosting fan interaction by 40%.
- Shipped an analytics dashboard (TypeScript, SQL, Palantir Foundry) reducing report generation time by 90%, with automated data refresh pipelines replacing manual reporting.

Acadia Analytics LLC Medford, MA
Quantitative Developer *Jan. 2025 – May 2025*
- Architected and deployed a full-stack trading service in Python/Flask on Linux, delivering real-time predictions to 50+ users at 99.5% uptime across a 4-month production window.
- Built a PyTest suite integrated into the deployment pipeline, reaching 85% code coverage across unit, integration, and end-to-end tests to catch regressions before production.

Tufts Human-Robot Interaction Lab Medford, MA
Research Assistant *Feb. 2024 – May 2025*
- Extended robot architecture with a Java + Redis caching module, reducing natural language processing latency by 46%; profiled call paths before and after optimization to quantify hotspot elimination.
- Built a real-time preprocessing service in Python, improving cognitive model input quality by 30%.

MIT Schwarzman College of Computing Cambridge, MA
Machine Learning Intern *Aug. 2024 – Dec. 2024*
- Developed feature extraction and model stacking workflows using **Scikit-learn, XGBoost** and **Random Forests** to predict harmful algae bloom risk; improved baseline accuracy by **32%**, reaching **76%** overall.
- Processed **11GB** of unstructured marine data using **NumPy, SQL,** and custom **ETL** scripts, reducing training time by **40%**.

Projects

MIT Lincoln Lab Capstone – Autonomous Vehicle Awareness | *PyTorch, Python, Matlab, Linux*
- Leading a team of 5 undergraduates building a 4D radar + vision perception stack with CNN and GAN models for real-time scene reconstruction on a Raspberry Pi running Linux.
- Built a Next.js + Flask observability dashboard streaming object detection output, scene visualization, and per-stage performance profiling metrics for live debugging.

DSCVR.AI Research App | *Next.js, Docker, AWS Bedrock*
- Built an academic research and writing platform with a fine-tuned LLM via Claude, deployed as Docker containers on AWS Bedrock with automated build and deploy workflows.
- Earned Cornell NYC AI Hackathon Winner and Most App Store Ready Award for seamless UX and ML integration.